

Endpoint crossings in the persistent random walk: uniform Bessel bounds and boundary effects

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Abstract

For a symmetric persistent random walk, we bound the relative error of a Bessel approximation to the ratio of each interior bin probability to an endpoint probability. If there are N steps and the switching odds are $u = (1 - p)/p$, the relative error is at most $Nu^2/2 + u$, whatever the bin. From this we get an expansion of the crossing point, with explicit constants and a $1/\log$ correction, that holds for all bins at a distance from the edge tending to infinity. For a bin at a fixed distance from the edge, the crossing comes instead from a finite Laguerre polynomial, and the next term turns out to be the same for all such bins. We also show that the two regimes match. We then prove the same kind of bound for periodic spin boundaries. The periodic and free central crossings differ in odds by $(\log 2)/N$ to leading order. In both cases the ratio near the center looks like $\exp(t - 2y^2)$, which tells us how many bins sit above the endpoints. The exact law, the normal limit and the Bessel scaling were known before, and so was the question of when the middle and the endpoints are equal. Our results are about how big the discrete error is when p and the bin change with N .

1 Introduction

Problem 131 of Kagey's Open Problem Collection [1] asks about a Galton board where the ball tends to keep going the same way. The first step is equally likely to be left or right, and after that each step repeats the one before it with probability p . In particular, it asks when the middle bin and the endpoints have equal probability. We study that comparison for every interior bin, including bins approaching an edge, and compare free and periodic spin boundaries.

Model and conventions. Let $P_N(k)$ denote the probability of exactly k right steps in N steps, with $P_0(0) = 1$. Write $q = 1 - p$, $\rho = 2p - 1$, and $u = q/p$ when $p > 0$. Thus $P_N(0) = P_N(N) = p^{N-1}/2$ for $N \geq 1$. Kagey's row n corresponds to $N = n - 1$. For $p = \alpha/(\alpha + \beta)$, his numerator tables are

$$n_N(k) = 2(\alpha + \beta)^{N-1} P_N(k) \quad (N \geq 1), \quad n_0(0) = 1.$$

These are the numerators as Kagey tabulates them, not reduced fractions.

Results and organization. Our main estimate, Theorem 6.2, bounds the relative error of a Bessel majorant independently of the bin. Its consequences are a crossing expansion uniform over every growing edge distance (Theorem 6.4), a fixed-edge Laguerre law and its matching limit (Theorems 6.5 and 6.6), and the periodic/free boundary shift and spatial critical window in

Section 7. We first give short proofs of the exact law, the lattice-path interpretations of Kagey's two numerator tables, and the variance and normal limit. Section 5 proves that the central crossing is unique and strictly increasing in N . The all-bin theorem also gives the central expansion, but we treat the center separately since the inversion is simplest there.

Prior work. The correlated random walk goes back to Goldstein [5] and Gillis [6]. Renshaw and Henderson [7] study the same symmetric model, motivated by a Galton machine, and give the exact law from run counts, the generating function, the normal limit, and the upturn at the endpoints. Their Section 4 also gives a Bessel limit at a fixed persistence scale, followed by a limit as that scale grows. Since the two limits are taken one after the other, they do not give an error bound at finite N when the scale grows like $\log N$, which is where the crossing happens. The run count used below is classical [8]. Dekking and Kong [10] give the Markov-binomial law, the variance, and a classification of the local modes. Their Proposition 4.3 proves interior log-concavity for an arbitrary initial distribution and $0 < a, b < 1$, and our model is the case $a = b = 1 - p$ with a symmetric initial law. We prove the ordering we need directly from the coefficients instead. The OEIS entries already record both tables [2, 3], and our bijections explain the connections Kagey points out.

Herrmann and Vallois [11] give the telegraph limit with its Bessel density. Antal, Droz and Rácz [12] treat the magnetization distribution at fixed scale under several boundary conditions, and Dantchev and Rudnick [13] give exact free and periodic coefficients. Stepanyan et al. [14, Section III.B] study the periodic endpoint-adjacent and endpoint-center crossings directly and give a finite-size approximation to the latter [14, Eq. (16)]. So we did not come up with the crossing question. Our part is bounding the discrete error when $Nu = O(\log N)$ (and near the edges), solving for the crossing from that bound, and finding the window of bins above the endpoints. The Markov-binomial estimates in [15, 16] assume ranges of the transition probabilities that exclude the symmetric limit $p \rightarrow 1$.

2 The exact distribution

A run of the experiment with $N \geq 1$ bounces is a word $w \in \{L, R\}^N$. Let $\text{chg}(w)$ be the number of indices $i < N$ with $w_i \neq w_{i+1}$ and $\text{rep}(w) = N - 1 - \text{chg}(w)$. By the definition of the model,

$$\mathbb{P}(w) = \frac{1}{2} p^{\text{rep}(w)} q^{\text{chg}(w)}. \quad (1)$$

Theorem 2.1 (bin probabilities). *Let $N \geq 1$. Then $P_N(0) = P_N(N) = \frac{1}{2}p^{N-1}$, and for $1 \leq k \leq N - 1$*

$$P_N(k) = \frac{1}{2} \sum_{j=1}^{N-1} W(N, k, j) p^{N-1-j} q^j,$$

where, writing $a = k$ and $b = N - k$,

$$W(N, k, 2i - 1) = 2 \binom{a-1}{i-1} \binom{b-1}{i-1}, \quad W(N, k, 2i) = \binom{a-1}{i} \binom{b-1}{i-1} + \binom{a-1}{i-1} \binom{b-1}{i}$$

is the number of words with a letters R , b letters L and exactly j direction changes.

Proof. By (1) it suffices to count words. A word with j changes consists of $j + 1$ maximal runs, alternately of R 's and of L 's. If $k \in \{0, N\}$ there is one word and it has no change. Let $a, b \geq 1$.

If $j + 1 = 2i$ there are i runs of each letter and the first letter is free. The run lengths form a composition of a into i parts and a composition of b into i parts, and there are $\binom{a-1}{i-1} \binom{b-1}{i-1}$ such pairs. If $j + 1 = 2i + 1$ the word starts and ends with the same letter. Starting with R gives $i + 1$ runs of R and i runs of L , so $\binom{a-1}{i} \binom{b-1}{i-1}$ words, and starting with L gives the other summand. $\square$

For instance $P_2 = (\frac{p}{2}, q, \frac{p}{2})$, $P_3(1) = P_3(2) = \frac{1}{2}(1 - p^2)$ and $P_4(2) = (1 - p)(1 - p + p^2)$.

Note that the only way to end in bin 0 is the word $L \cdots L$, which never switches. Every other bin needs at least one switch. So when p is close to 1 the endpoints win, and the question in Section 5 is how close p has to be.

Proposition 2.2 (recurrence and generating function). *Let $R_N(k)$ and $L_N(k)$ be the probabilities of being in bin k after N bounces with last bounce R , respectively L . Then $R_1(1) = L_1(0) = \frac{1}{2}$ and*

$$R_N(k) = p R_{N-1}(k-1) + q L_{N-1}(k-1), \quad L_N(k) = p L_{N-1}(k) + q R_{N-1}(k). \quad (2)$$

Consequently

$$\sum_{N \geq 0} \sum_{k=0}^N P_N(k) x^k t^N = \frac{1 - (p - \frac{1}{2})(1+x)t}{1 - p(1+x)t + (2p-1)xt^2}, \quad (3)$$

and for all $N \geq 2$ and all k (with $P_M(k) = 0$ outside $0 \leq k \leq M$)

$$P_N(k) = p(P_{N-1}(k) + P_{N-1}(k-1)) - (2p-1)P_{N-2}(k-1). \quad (4)$$

Proof. The recursion (2) follows by conditioning on the last two bounces. Put $F_R = \sum R_N(k)x^k t^N$ and $F_L = \sum L_N(k)x^k t^N$. Then (2) and the initial values give $F_R = xt(\frac{1}{2} + pF_R + qF_L)$ and $F_L = t(\frac{1}{2} + pF_L + qF_R)$. The determinant of this linear system is $(1 - pxt)(1 - pt) - q^2xt^2 = 1 - p(1+x)t + (p^2 - q^2)xt^2$, and $p^2 - q^2 = 2p - 1$. Cramer's rule gives $F_R = \frac{1}{2}xt(1 - pt)/\Delta$ and $F_L = \frac{1}{2}t(1 - pxt)/\Delta$ with Δ the determinant, and $1 + F_R + F_L$ simplifies to (3). Multiplying (3) by Δ and reading off the coefficient of $x^k t^N$ for $N \geq 2$ gives (4). $\square$

At $p = \frac{1}{2}$ the last term of (4) vanishes and (4) is Pascal's rule with weights $\frac{1}{2}$. In terms of numerators, (4) reads $n_N(k) = \alpha(n_{N-1}(k) + n_{N-1}(k-1)) - (\alpha^2 - \beta^2)n_{N-2}(k-1)$ for $N \geq 3$.

3 The numerator tables and lattice paths

Proposition 3.1. *Let $p = \alpha/(\alpha + \beta)$ with integers $\alpha, \beta \geq 0$, not both zero. For $N \geq 1$,*

$$n_N(k) = \sum_w \alpha^{\text{rep}(w)} \beta^{\text{chg}(w)},$$

the sum over the words with k letters R and $N - k$ letters L . In particular the numerators are integers, and

$$\sum_{a,b \geq 0} n_{a+b}(a) x^a y^b = \frac{1 - (\alpha - 1)(x + y) + (\alpha - \beta)(\alpha + \beta - 2)xy}{1 - \alpha(x + y) + (\alpha^2 - \beta^2)xy}.$$

Proof. Multiply (1) by $2(\alpha + \beta)^{N-1}$. For the generating function, let F_R and F_L be the sums of the weighted words ending in R and in L . Then $F_R = x(1 + \alpha F_R + \beta F_L)$ and $F_L = y(1 + \alpha F_L + \beta F_R)$, and solving as in Proposition 2.2 gives the stated value of $1 + F_R + F_L$. $\square$

For $(\alpha, \beta) = (2, 1)$ the generating function is $(1 - x)(1 - y)/(1 - 2x - 2y + 3xy)$, and for $(\alpha, \beta) = (1, 2)$ it is $(1 - xy)/(1 - x - y - 3xy)$. These are the generating functions recorded in A035002 (by Kruchinin and by Somos) and in A348595 (by Mathar) [2, 3], which already identifies the tables. We prove the identifications directly from the definitions of the two sequences, which shows where the weights 2 come from.

Theorem 3.2 ($p = \frac{2}{3}$ and rook paths). *Let $T(a, b)$ be the number of ways to move a rook from $(0, 0)$ to (a, b) by moves that increase one coordinate by a positive amount. Then for $p = \frac{2}{3}$, $n_N(k) = T(k, N - k)$ for all $N \geq 0$ and $0 \leq k \leq N$. In the indexing of A035002, whose array starts at $(1, 1)$, this is the entry $(k + 1, N - k + 1)$, so Kagey's $p = \frac{2}{3}$ table read by rows is A035002 read by antidiagonals.*

Proof. A035002 is defined by the rule that each entry is the sum of all entries to its left and all entries above it, which is the recursion for T obtained by removing the last rook move (the OEIS entry records this interpretation). Given a rook path to (a, b) with $a + b = N \geq 1$, replace each move of length ℓ to the right by R^ℓ and each move of length ℓ upward by L^ℓ . This gives a word w with a letters R and b letters L , together with the set of positions $i < N$ at which one move ends and the next begins. Such a cut is forced wherever $w_i \neq w_{i+1}$ and is optional wherever $w_i = w_{i+1}$, and every choice of optional cuts comes from exactly one rook path. Hence, $T(a, b) = \sum_w 2^{\text{rep}(w)}$, which is $n_N(a)$ for $(\alpha, \beta) = (2, 1)$ by Proposition 3.1. $\square$

Theorem 3.3 ($p = \frac{1}{3}$ and Mathar's blocked walks). *Call a lattice point (x, y) blocked if $x \equiv y \equiv 1$ or $x \equiv y \equiv 2 \pmod{3}$. Let $M(a, b)$ be the number of walks from $(0, 0)$ to $(3a, 3b)$ with unit steps right and up that visit no blocked point. Then for $p = \frac{1}{3}$, $n_N(k) = M(k, N - k)$. The array M is symmetric, and A348595 lists its lower triangle $M(n, k)$, $0 \leq k \leq n$. This is the sense in which that entry is "related" to the $p = \frac{1}{3}$ table.*

Proof. Reduce coordinates modulo 3. The allowed residues are $(0, 0)$, $(1, 0)$, $(2, 0)$, $(2, 1)$, $(0, 1)$, $(0, 2)$, $(1, 2)$. We call a point with residue $(0, 0)$ a *corner*. Let X be the number of times the walk steps onto a vertical line $x \equiv 0$, and Y the number of times it steps onto a horizontal line $y \equiv 0$. From a corner the walk goes right or up. After a right step it is at residue $(1, 0)$, and up is blocked, so it goes on to $(2, 0)$. Call this state A. From A, a right step lands on a corner and increases X . An up step leads to $(2, 1)$, where up is blocked, so the walk goes right to $(0, 1)$, which increases X , and then up to $(0, 2)$ since right is blocked. Call this state B. By symmetry, an up step from a corner leads to B without changing X or Y , and from B the walk reaches either a corner or A, with Y going up by one either way.

So each visit to A gives one letter R (an increase of X) and is followed by a corner or by B. Each visit to B gives one letter L and is followed by a corner or by A. A corner leads to A or B and gives no letter. Hence a walk from the corner $(0, 0)$ to the corner $(3a, 3b)$ gives a word w with a letters R and b letters L , where w_i comes from a visit to A if it is R and to B if it is L . Between w_i and w_{i+1} the walk goes from one of these states to the next. If $w_i = w_{i+1}$ it has to pass through a corner, so there is one way. If $w_i \neq w_{i+1}$ it can pass through a corner or go directly, so there are two ways. The start and the end are forced. Hence $M(a, b) = \sum_w 2^{\text{chg}(w)}$, which is $n_N(a)$ for $(\alpha, \beta) = (1, 2)$. In Mathar's note [4] attached to A348595, the "squares" in the name of the sequence are the lattice points (x, y) themselves, as his recurrences show, so M is his count. Counting the walks directly also gives the published terms (Section 8). $\square$

Figure 1 shows both bijections for the word $RRLRL$.

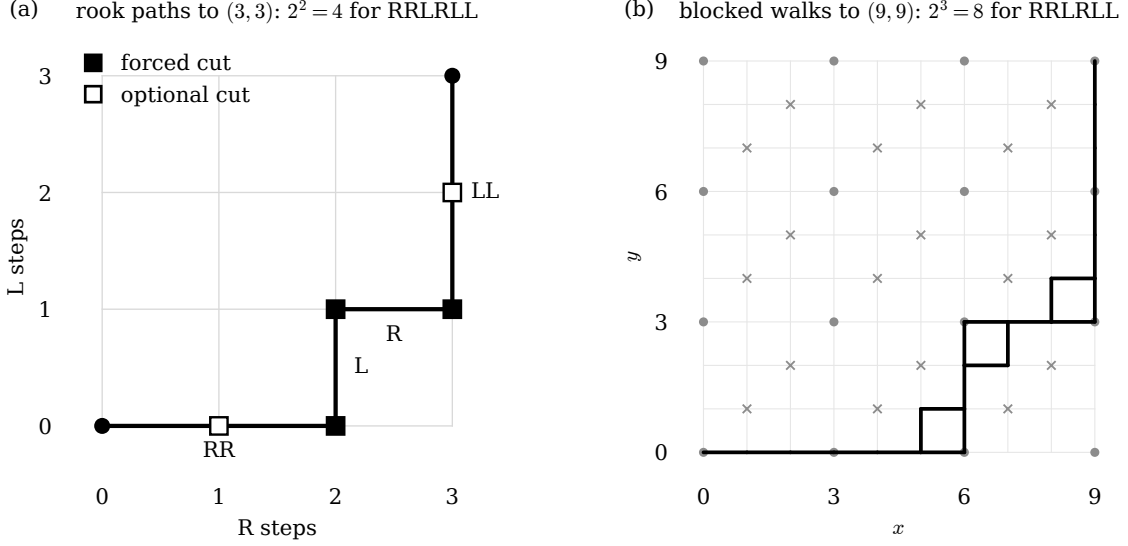


Figure 1: The two bijections for the word $RRLRL$. (a) The rook path. A cut is forced at each of the 3 direction changes (filled) and optional between equal letters (hollow), so the word comes from $2^2 = 4$ rook paths to $(3, 3)$. (b) The union of the $2^3 = 8$ blocked walks to $(9, 9)$ that give the word, with one detour square for each direction change. Blocked points are marked $\times$ and corners by dots.

For general $p = \alpha/(\alpha + \beta)$, Proposition 3.1 says that $n_N(k)$ counts monotone lattice paths from $(0, 0)$ to $(k, N - k)$ in which each straight continuation is colored in one of α ways and each corner in one of β ways. This gives the general weighted lattice-path interpretation.

4 Variance and the normal limit

Write $e_i = +1$ if the i -th bounce is R and $e_i = -1$ otherwise, $X_N = e_1 + \dots + e_N$, and $K_N = (N + X_N)/2$ for the bin index. Then $e_{i+1} = e_i s_i$, where $s_1, s_2, \dots$ are independent of each other and of e_1 , with $\mathbb{P}(s_i = 1) = p$ and $\mathbb{P}(s_i = -1) = q$, so $\mathbb{E}s_i = \rho$.

Proposition 4.1 (mean and variance). $\mathbb{E}K_N = N/2$, and for $p < 1$

$$\text{Var } K_N = \frac{1}{4} \left(N \frac{1 + \rho}{1 - \rho} - \frac{2\rho(1 - \rho^N)}{(1 - \rho)^2} \right) = \frac{Np}{4(1 - p)} - \frac{(2p - 1)(1 - (2p - 1)^N)}{8(1 - p)^2}.$$

For $p = 1$, $\text{Var } K_N = N^2/4$.

Proof. Note that $\mathbb{E}e_1 = 0$ and $e_{i+m} = e_i s_i \dots s_{i+m-1}$, so $\mathbb{E}e_i = 0$ and $\mathbb{E}e_i e_{i+m} = \rho^m$. Hence, $\text{Var } X_N = N + 2 \sum_{m=1}^{N-1} (N - m) \rho^m$. Summing the two geometric series gives the first form, and the second follows from $(1 + \rho)/(1 - \rho) = p/q$ and $1 - \rho = 2q$. $\square$

Theorem 4.2 (normal limit). Let $0 < p < 1$. Then $(K_N - N/2)/\sqrt{N}$ converges in distribution to the normal law with mean 0 and variance $p/(4(1 - p))$.

Proof. Let $\varphi_N(\theta) = \mathbb{E}e^{i\theta X_N}$ and let v_N be the column vector with entries $\mathbb{E}[e^{i\theta X_N}; e_N = +1]$ and $\mathbb{E}[e^{i\theta X_N}; e_N = -1]$. By (2), $v_{N+1} = M v_N$ with

$$M = \begin{pmatrix} p e^{i\theta} & q e^{i\theta} \\ q e^{-i\theta} & p e^{-i\theta} \end{pmatrix}, \quad \text{tr } M = 2p \cos \theta, \quad \det M = p^2 - q^2 = \rho.$$

The eigenvalues are $\lambda_{\pm}(\theta) = p \cos \theta \pm \sqrt{q^2 - p^2 \sin^2 \theta}$, using $p^2 - \rho = q^2$. For $|\sin \theta| < q/p$ they are real and distinct, so $\varphi_N = c_+ \lambda_+^{N-1} + c_- \lambda_-^{N-1}$ for $N \geq 1$, with $c_{\pm}(\theta)$ determined by $\varphi_1 = \cos \theta$ and $\varphi_2 = p \cos 2\theta + q$, which give

$$c_+ = \frac{\varphi_2 - \lambda_- \cos \theta}{\lambda_+ - \lambda_-}, \quad c_- = \cos \theta - c_+.$$

As $\theta \rightarrow 0$ we have $\lambda_+ \rightarrow 1$, $\lambda_- \rightarrow \rho$, $c_+ \rightarrow 1$, $c_- \rightarrow 0$, and

$$\lambda_+(\theta) = p(1 - \frac{\theta^2}{2}) + q(1 - \frac{p^2 \theta^2}{2q^2}) + O(\theta^4) = 1 - \frac{p}{2q} \theta^2 + O(\theta^4).$$

Fix $s \in \mathbb{R}$ and put $\theta = s/\sqrt{N}$. Since $|\rho| < 1$, $c_- \lambda_-^{N-1} \rightarrow 0$, and $\lambda_+^{N-1} \rightarrow \exp(-\frac{p}{2q} s^2)$. Hence, $\varphi_N(s/\sqrt{N}) \rightarrow \exp(-\frac{p}{2q} s^2)$, and by Lévy's continuity theorem [9, Theorem 26.3] $X_N/\sqrt{N}$ converges to the normal law with variance p/q . Finally $K_N - N/2 = X_N/2$. $\square$

For $p = 1$ the ball is in bin 0 or bin N with probability $\frac{1}{2}$ each, and for $p = 0$ it is in a middle bin, so there is no normal limit in these two cases.

Remark 4.3 (cylinder). On a cylinder whose rows have c pegs the local rule is unchanged, so the position is X_N modulo $2c$ (in half-peg units), and the discrete Fourier coefficients of its law are $\varphi_N(\pi l/c)$, $0 \leq l < 2c$. The formula $\varphi_N = c_+ \lambda_+^{N-1} + c_- \lambda_-^{N-1}$ holds whenever $\lambda_+ \neq \lambda_-$. For $0 < p < 1$ and $\theta \notin \pi \mathbb{Z}$ both eigenvalues have modulus less than 1. Real eigenvalues satisfy $|\lambda_{\pm}| \leq p|\cos \theta| + q$, and complex ones have modulus $\sqrt{\rho}$. The coefficients at $l = 0$ and $l = c$ are 1 and $(-1)^N$. Hence, on the cylinder the law approaches the uniform law on the c positions of the correct parity geometrically fast, and there is no normal shape in the limit.

5 When does the middle equal the extremes?

Let $N \geq 2$ and let $m_N = \lfloor N/2 \rfloor$ be a middle bin (for odd N the two middle bins have equal probability by symmetry). Put $u = q/p$. By Theorem 2.1,

$$\frac{P_N(m_N)}{P_N(0)} = S_N(u) := \sum_{j=1}^{N-1} W(N, m_N, j) u^j, \quad (5)$$

a polynomial in u with nonnegative integer coefficients and $W(N, m_N, 1) = 2$.

We expect $1 - p_N$ to be about $(\log N)/N$. Each switch in a word gives a factor of u , and a word that ends in the middle can switch almost anywhere, so adding them up gives roughly e^{Nu}/N . This is 1 when $Nu \approx \log N$. Theorem 5.2 proves this with the correction terms.

Theorem 5.1. (a) For every $N \geq 2$ there is exactly one $p = p_N \in [0, 1]$ with $P_N(m_N) = P_N(0)$. The middle is more likely than the extremes for $p < p_N$ and less likely for $p > p_N$.

(b) $p_2 = \frac{2}{3}$, $p_3 = 1/\sqrt{2}$, p_4 is the real root of $3p^3 - 4p^2 + 4p - 2$, and $p_N < p_{N+1}$ for all $N \geq 2$.

(c) $p_N \rightarrow 1$, and more precisely $1 - p_N \sim (\log N)/N$.

Proof. (a) At $p = 0$ the extremes have probability 0 and the middle does not. For $p > 0$ the equation is $S_N(u) = 1$ by (5). S_N is continuous and strictly increasing on $[0, \infty)$ with $S_N(0) = 0$ and $S_N(u) \geq 2u$, so there is exactly one root $u_N \in (0, \frac{1}{2}]$, and $p = 1/(1 + u)$ is a decreasing bijection from $[0, \infty)$ to $(0, 1]$.

(b) $S_2 = 2u$, $S_3 = 2u + u^2$ and $S_4 = 2u + 2u^2 + 2u^3$ give the three values. For p_4 , substitute $u = (1 - p)/p$ in $S_4(u) = 1$. For monotonicity we compare coefficients. If $N = 2m$ then the middle $(a, b) = (m, m)$ becomes $(m, m + 1)$ for $N + 1$, and if $N = 2m + 1$ then $(m, m + 1)$ becomes $(m + 1, m + 1)$. In both cases one of a, b grows by one and the other is unchanged, the formulas of Theorem 2.1 are symmetric in (a, b) , and every binomial coefficient $\binom{a-1}{\cdot}$ is nondecreasing in a . So $W(N + 1, m_{N+1}, j) \geq W(N, m_N, j)$ for every j . The coefficient of u^2 is $a + b - 2 = N - 2$, which increases strictly. Hence, $S_{N+1}(u) > S_N(u)$ for $u > 0$, so $u_{N+1} < u_N$.

(c) Let $a = \lfloor N/2 \rfloor$ and $b = \lceil N/2 \rceil$, and fix $\varepsilon \in (0, \frac{1}{2})$.

Lower bound for u_N . Using $\binom{a-1}{r}, \binom{b-1}{r} \leq b^r/r!$ and writing $x = bu$, the terms of $S_N(u)$ with $j = 2i - 1$ are at most $\frac{2}{b} x(x^{i-1}/(i-1)!)^2$ and those with $j = 2i$ are at most $\frac{2}{b} \frac{x^i}{i!} \frac{x^i}{(i-1)!}$. Since $\sum_r (x^r/r!)^2 \leq e^{2x}$ and $\sum_i \frac{x^i}{i!} \frac{x^i}{(i-1)!} \leq e^x \cdot x e^x$,

$$S_N(u) \leq \frac{4x}{b} e^{2x}.$$

If $x \leq (\frac{1}{2} - \varepsilon) \log b$ the right side is at most $2b^{-2\varepsilon} \log b$, which is less than 1 for large b . Hence, $b u_N > (\frac{1}{2} - \varepsilon) \log b$ for large N .

Upper bound for u_N . Let $u^* = (\frac{1}{2} + \varepsilon)(\log a)/a$, $y = au^*$ and $r = \lfloor y \rfloor$, so $r \geq 1$ for large N . Keep only the term $j = 2r + 1$ of $S_N(u^*)$ and use $\binom{b-1}{r} \geq \binom{a-1}{r} \geq (a - r)^r/r!$ to get

$$S_N(u^*) \geq 2u^* \left(\frac{(a - r)^r (u^*)^r}{r!} \right)^2 = \frac{2y}{a} \left(1 - \frac{r}{a} \right)^{2r} \left(\frac{y^r}{r!} \right)^2.$$

From $\log r! \leq \int_1^r \log t \, dt + \log r$ we get $r! \leq e r (r/e)^r$, so $y^r/r! \geq r^r/r! \geq e^{r-1}/r \geq e^{y-2}/y$. Also $(1 - r/a)^{2r} \geq 1 - 2r^2/a \rightarrow 1$. Therefore

$$S_N(u^*) \geq (1 - o(1)) \frac{2e^{2y-4}}{ay} = (1 - o(1)) \frac{2e^{-4}a^{2\varepsilon}}{y} \rightarrow \infty,$$

so $S_N(u^*) > 1$ and $u_N < u^*$ for large N .

Since $a, b = N/2 + O(1)$, the two bounds give $u_N \sim \log(N/2)/N \sim (\log N)/N$, and $1 - p_N = u_N/(1 + u_N) \sim u_N$. $\square$

In Kagey's terms, for the $(2n + 1)$ -st row ($N = 2n$) the unique value is p_{2n} , and the least of these is $p_2 = \frac{2}{3}$, which is why the third row of his $p = \frac{2}{3}$ table is 2, 2, 2. For the $(2n)$ -th row ($N = 2n - 1, n \geq 2$) the least is $p_3 = 1/\sqrt{2}$. The second row is 1, 1 for every p . There is no greatest value, since $p_N \uparrow 1$. The tenth row has $p_9 = 0.82829 \dots$, just below $\frac{5}{6}$, which is why the extremes and the middle are nearly level in his third histogram. Figure 2 shows this row at the four values of p in his histograms.

N	p_N	N	p_N	N	p_N	N	p_N
2	0.66666667	6	0.78871211	10	0.83840411	50	0.94200405
3	0.70710678	7	0.80361731	11	0.84652725	100	0.96496633
4	0.74487450	8	0.81759938	12	0.85426060	200	0.97939018
5	0.76759188	9	0.82829163	20	0.89293837	400	0.98812464

Table 1: The value p_N at which the middle and extreme bins of row $N + 1$ are equally likely.

The Bessel functions come in because the runs are long. When Nu is about $\log N$, a typical word only switches about $\log N$ times. Then the number of ways to split a letters into r runs,

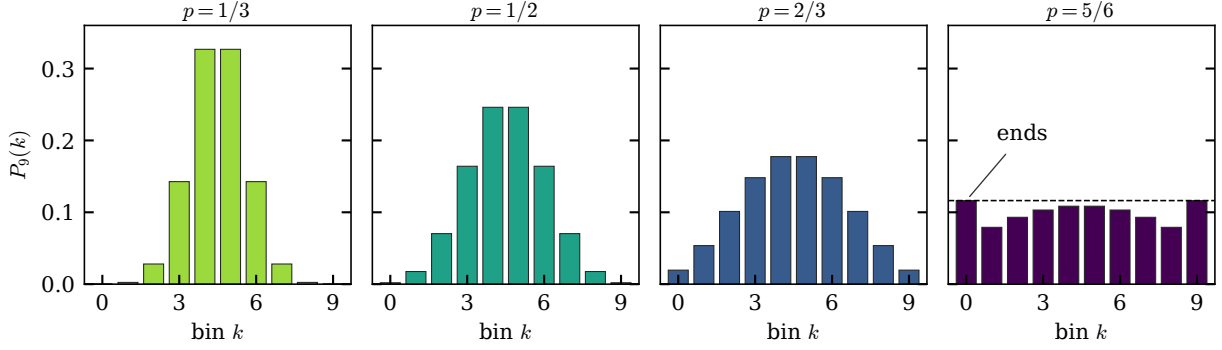


Figure 2: The bin probabilities $P_9(k)$ of Kagey's tenth row. At $p = 5/6$, just above p_9 , the two middle bins are slightly below the ends (dashed line).

$\binom{a-1}{r}$, is close to $a^r/r!$, and after this change the run count is exactly the power series of I_0 and I_1 . The next theorem bounds the error we make by doing this.

Theorem 5.2 (uniform Bessel approximation). *Put $h = N/2$, $J(x) = I_0(2x) + I_1(2x)$ and $F(x) = 2xJ(x)$. For every $N \geq 2$ and $x > 0$,*

$$0 \leq 1 - \frac{hS_N(x/h)}{F(x)} \leq \frac{x(x+1)}{h}. \quad (6)$$

Let x_B be the unique positive root of $F(x_B) = h$, and put $u_B = x_B/h$, $u_N = (1 - p_N)/p_N$. Then

$$0 \leq u_N - u_B = O((\log N)^2/N^2).$$

Writing $L = \log N$ and $c_0 = \frac{1}{2} \log(\pi/8)$, we have

$$N(1 - p_N) = L - \frac{1}{2} \log L + c_0 + O\left(\frac{\log L}{L}\right), \quad (7)$$

$$N(1 - p_N) = L - \frac{1}{2} \log L + c_0 + \frac{\frac{1}{4} \log L - \frac{1}{2} c_0 + \frac{1}{8}}{L} + O\left(\frac{(\log L)^2}{L^2}\right). \quad (8)$$

Proof. Let $a = \lfloor N/2 \rfloor$, $b = \lceil N/2 \rceil$, and define

$$A_r = \frac{r!}{h^r} \binom{a-1}{r}, \quad B_r = \frac{r!}{h^r} \binom{b-1}{r} \quad (r \geq 0).$$

With $(v)_+ = \max(v, 0)$, these have the product expressions

$$A_r = \prod_{k=1}^r \left(1 - \frac{h-a+k}{h}\right)_+, \quad B_r = \prod_{k=1}^r \left(1 - \frac{h-b+k}{h}\right)_+.$$

They lie in $[0, 1]$, including beyond their finite support, where a factor is zero. For nonnegative t_j , the inequality $1 - \prod_j (1 - t_j)_+ \leq \sum_j t_j$ holds even when some $t_j > 1$. Since $a + b = 2h$, it gives

$$0 \leq 1 - A_r B_r \leq \frac{r(r+1)}{h}.$$

The deficits for $A_{r+1}B_r$ and A_rB_{r+1} are at most $((r+1)^2 + h - a)/h$ and $((r+1)^2 + h - b)/h$, respectively. Averaging cancels the parity offsets, so

$$0 \leq 1 - \frac{A_{r+1}B_r + A_rB_{r+1}}{2} \leq \frac{(r+1)^2}{h}.$$

The run-count formula, with $u = x/h$, becomes

$$S_N(u) = 2u \left[\sum_{r \geq 0} \frac{x^{2r}}{(r!)^2} A_r B_r + \sum_{r \geq 0} \frac{x^{2r+1}}{r!(r+1)!} \frac{A_{r+1}B_r + A_rB_{r+1}}{2} \right].$$

Dropping the A, B factors yields $2uJ(x) = F(x)/h$, using the Bessel series [17, Eq. 10.25.2]. Index shifts in those absolutely convergent series give

$$\begin{aligned} \sum_{r \geq 0} r(r+1) \frac{x^{2r}}{(r!)^2} &= x^2 I_0(2x) + x I_1(2x), \\ \sum_{r \geq 0} (r+1)^2 \frac{x^{2r+1}}{r!(r+1)!} &= x^2 I_1(2x) + x I_0(2x). \end{aligned}$$

Their sum is $x(x+1)J(x)$, proving (6), including the tail beyond the finite polynomial support. In particular the relative error is $O((\log N)^2/N)$ uniformly for $0 < x \leq C \log N$ for each fixed C .

Now let $x_N = hu_N$. Theorem 5.1 gives $x_N \sim \frac{1}{2} \log N$. Differentiation gives $F'(x) = 2F(x) + 2I_0(2x)$, hence $F'/F \geq 2$. Set $\delta_N = x_N(x_N + 1)/h$. For large N , $\delta_N < 1$, and (6) at the root gives

$$h \leq F(x_N) \leq \frac{h}{1 - \delta_N}, \quad 0 \leq x_N - x_B \leq -\frac{1}{2} \log(1 - \delta_N) = O((\log N)^2/N).$$

Divide by h for the assertion about u . Moreover,

$$N(1 - p_N) = \frac{2x_N}{1 + x_N/h} = 2x_B + O((\log N)^2/N).$$

Put $z = 2x_B$. The standard large-argument expansion [17, Eq. 10.40.1] is

$$I_0(z) + I_1(z) = \frac{2e^z}{\sqrt{2\pi z}} \left(1 - \frac{1}{8z} + O(z^{-2}) \right).$$

Since $z(I_0(z) + I_1(z)) = N/2$, taking logarithms gives

$$z + \frac{1}{2} \log z = L + c_0 + \frac{1}{8z} + O(z^{-2}).$$

First $z = L + O(\log L)$, then substitution proves (7). For one further term put $z = L + d + e/L$, where $d = -\frac{1}{2} \log L + c_0$. Matching coefficients of $1/L$ gives $e = \frac{1}{8} - d/2$. The residual is $O((\log L)^2/L^2)$. The derivative of $z + \frac{1}{2} \log z$ is at least one, so inverting keeps that order. This proves (8). $\square$

The ratios u_B/u_N are 0.9761, 0.9864, 0.9924 at $N = 100, 200, 400$ and 0.99579, 0.99769, 0.99874 at $N = 800, 1600, 3200$. We predicted intervals for the last three before computing them (data/ledger.md), and Theorem 5.2 proves the approximation behind those predictions. Figure 3 compares the exact roots with these approximations up to $N = 10^4$.

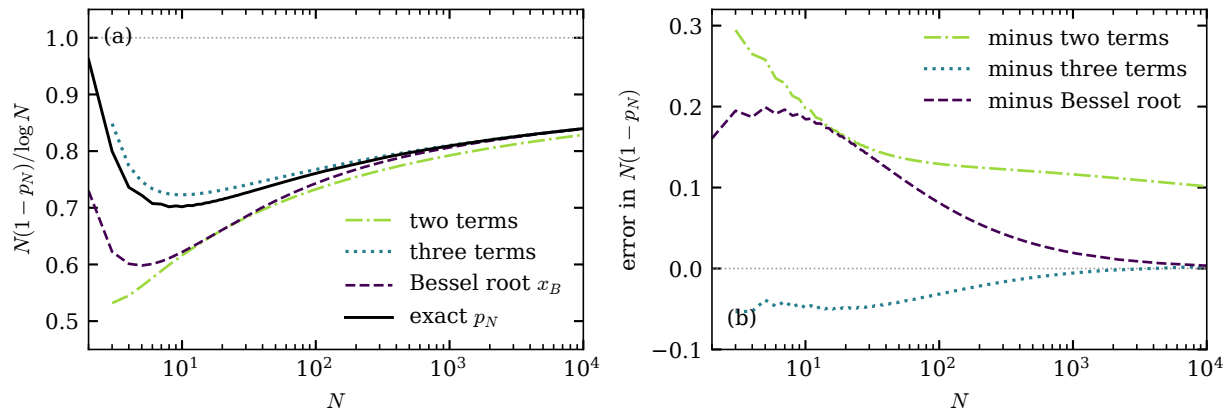


Figure 3: (a) The scaled crossing $N(1 - p_N)/\log N$ from the exact roots, from the Bessel root u_B , and from the expansions (7) and (8) without their error terms. It tends to 1 very slowly. (b) The error of each approximation to $N(1 - p_N)$.

Proposition 5.3 (adjacent bins and global extrema). *For $N \geq 2$ and $0 < p < 1$, an adjacent bin has the same probability as an endpoint precisely at*

$$p_N^{\text{adj}} = \frac{1 + \sqrt{N-1}}{2 + \sqrt{N-1}}.$$

For $0 < p < 1$, the endpoints are the least likely bins if $p < p_N^{\text{adj}}$, bins 1 and $N-1$ are if $p > p_N^{\text{adj}}$, and they tie at equality. For $N \geq 4$, the central crossing satisfies $p_N > p_N^{\text{adj}}$. At $p = p_N$, the middle and endpoints attain the global maximum and bins $1, N-1$ attain the global minimum. For $N = 2, 3$, $p_N = p_N^{\text{adj}}$ and every bin ties at the crossing.

Proof. A word with one R has one change in two cases and two changes in $N-2$ cases. Thus $P_N(1)/P_N(0) = 2u + (N-2)u^2$, whose positive root at level one is $u^{\text{adj}} = 1/(1 + \sqrt{N-1})$, including $N = 2$. This yields the stated p and sign change. By Proposition 6.1 in Section 6, which we prove from the run-count formula, the interior bins get more likely as we move toward the middle. By symmetry, the middle is then the largest interior bin and bins 1 and $N-1$ are the smallest. Comparing bins $1, N-1$ with the endpoints gives the claims about the global minimum.

Put $a = \lfloor N/2 \rfloor$ and $b = \lceil N/2 \rceil$. The polynomial $S_N(u)$ begins with $2u + (N-2)u^2$ and has nonnegative coefficients. For $N \geq 4$, its cubic coefficient is $2(a-1)(b-1) > 0$. Consequently, at its positive root u_N ,

$$2u_N + (N-2)u_N^2 < S_N(u_N) = 1.$$

The adjacent polynomial is strictly increasing, so $u_N < u^{\text{adj}}$ and hence $p_N > p_N^{\text{adj}}$. The middle and endpoints tie by definition of p_N , proving the global-maximum assertion as well. For $N = 2, 3$, $S_N(u) = 2u + (N-2)u^2$, and symmetry gives the stated ties. In particular $p_3 = p_3^{\text{adj}} = 1/\sqrt{2}$. $\square$

The adjacent-bin scale is $1 - p_N^{\text{adj}} \sim N^{-1/2}$, distinct from $1 - p_N \sim (\log N)/N$. This makes sense. Bin 1 has a single R , and unless it is at one of the ends it needs two switches. So $P_N(1)/P_N(0) \approx Nu^2$, which only reaches 1 at $u \approx N^{-1/2}$. The middle bin has far more words, so it catches up to the endpoints much earlier. At $p = 0, 1$ there are also ties between bins of probability zero.

6 The full hierarchy of endpoint crossings

The middle and adjacent bins are the two ends of a larger family of crossing thresholds. In this section, a, b are positive integers, $N = a + b$, and

$$S_{a,b}(u) = \frac{P_{a+b}(a)}{P_{a+b}(0)}, \quad u = \frac{1-p}{p}.$$

This ratio is defined for $p > 0$. The run-count formula can be written as

$$S_{a,b}(u) = \sum_{r \geq 0} \binom{a-1}{r} \binom{b-1}{r} \left(2u^{2r+1} + \frac{N-2-2r}{r+1} u^{2r+2} \right). \quad (9)$$

Indeed, the even coefficient follows from

$$\binom{a-1}{r+1} \binom{b-1}{r} + \binom{a-1}{r} \binom{b-1}{r+1} = \frac{N-2-2r}{r+1} \binom{a-1}{r} \binom{b-1}{r}.$$

The summation is finite and all its coefficients are nonnegative. Since the linear coefficient is 2, there is a unique $u_{a,b} > 0$ satisfying $S_{a,b}(u_{a,b}) = 1$. Write

$$p_{a,b} = \frac{1}{1 + u_{a,b}}.$$

Thus every interior bin has exactly one endpoint crossing in $0 < p < 1$. We exclude $p = 0$, where a noncentral bin and an endpoint can both have probability zero.

Proposition 6.1 (ordering the thresholds). *If $a < b - 1$, then*

$$u_{a+1,b-1} < u_{a,b}, \quad p_{a+1,b-1} > p_{a,b}.$$

Also $p_{a,b+1} > p_{a,b}$ for all positive a, b . In fact, the corresponding inequalities between the polynomials $S_{a+1,b-1}$ and $S_{a,b}$, or $S_{a,b+1}$ and $S_{a,b}$, hold coefficientwise, with at least one strict coefficient.

Proof. For fixed N and $r \leq a - 1$, write

$$\binom{a-1}{r} \binom{b-1}{r} = \frac{1}{(r!)^2} \prod_{j=0}^{r-1} (a-1-j)(b-1-j).$$

Every factor increases after (a, b) is replaced by $(a+1, b-1)$, since

$$(a-j)(b-2-j) - (a-1-j)(b-1-j) = b-a-1 > 0.$$

Any new summand is nonnegative, and the cubic coefficient increases strictly. Equation (9) therefore gives the first assertion. Increasing b with a fixed increases the binomial coefficients in the original run formula, and increases the quadratic coefficient $N - 2$ strictly. Comparing the unique roots proves the remaining assertions. $\square$

Relation to interior log-concavity. The ordering of the bin heights also follows from the interior log-concavity proved by Dekking and Kong [10, Proposition 4.3]. We use the coefficient argument above instead, in Proposition 5.3. The results below locate all the crossings, including those of bins whose distance from the edge changes with N .

6.1 A Bessel estimate uniform in the bin

In bin a , there are about $(a^r/r!)(b^r/r!)$ words with r runs of each letter, and each has weight about u^{2r} . Together this only depends on abu^2 , so the right variable is $\sqrt{ab}u$. In other words a bin near the edge acts like the middle of a shorter row, with $\sqrt{ab}$ in place of $N/2$.

Put

$$s = \sqrt{ab}, \quad A = \frac{ab}{a+b}, \quad c = \frac{a+b}{2\sqrt{ab}}, \quad \eta = \frac{1}{c} = \frac{2\sqrt{ab}}{a+b}, \quad x = su.$$

Define

$$J_c(x) = I_0(2x) + cI_1(2x), \quad T_{a,b}(u) = 2uJ_c(su).$$

Theorem 6.2 (uniform estimate for every bin). *For all positive integers a, b and all $u > 0$,*

$$0 \leq 1 - \frac{S_{a,b}(u)}{T_{a,b}(u)} \leq \frac{(a+b)u^2}{2} + u = \frac{x^2}{2A} + \frac{x}{s}. \quad (10)$$

In particular, for fixed N and u the bound does not depend on the bin. The inequality still holds when the right side exceeds one, but then it gives no lower bound on $S_{a,b}$.

Proof. Set

$$U_r = \frac{r!}{a^r} \binom{a-1}{r}, \quad V_r = \frac{r!}{b^r} \binom{b-1}{r}, \quad H = \frac{1}{a} + \frac{1}{b}.$$

These numbers lie in $[0, 1]$. Their product representations, using positive parts beyond the finite support, and $1 - \prod_i (1 - t_i)_+ \leq \sum_i t_i$ for $t_i \geq 0$, give

$$0 \leq 1 - U_r V_r \leq \frac{H}{2} r(r+1). \quad (11)$$

For the even terms define

$$E_r = \frac{aU_{r+1}V_r + bU_rV_{r+1}}{a+b}.$$

The same argument, with the indicated weights, gives

$$\begin{aligned} 0 \leq 1 - E_r &\leq \frac{(r+1)(r+2)}{a+b} + \frac{r(r+1)}{2(a+b)} \left(\frac{a}{b} + \frac{b}{a} \right) \\ &= \frac{H}{2} r(r+1) + \frac{2(r+1)}{a+b}. \end{aligned} \quad (12)$$

The exact polynomial is

$$S_{a,b}(u) = 2u \left[\sum_{r \geq 0} \frac{x^{2r}}{(r!)^2} U_r V_r + c \sum_{r \geq 0} \frac{x^{2r+1}}{r!(r+1)!} E_r \right]. \quad (13)$$

Dropping the product factors gives $T_{a,b}$. Index shifts in the Bessel series yield

$$\begin{aligned} \sum_{r \geq 0} r(r+1) \frac{x^{2r}}{(r!)^2} &= x^2 I_0(2x) + x I_1(2x), \\ \sum_{r \geq 0} r(r+1) \frac{x^{2r+1}}{r!(r+1)!} &= x^2 I_1(2x), \\ \sum_{r \geq 0} (r+1) \frac{x^{2r+1}}{r!(r+1)!} &= x I_0(2x). \end{aligned}$$

Consequently the error in (13), after division by $2u$, is at most

$$\frac{H}{2}x^2J_c(x) + x\left\{\frac{H}{2}I_1(2x) + \frac{2c}{a+b}I_0(2x)\right\} = \left(\frac{x^2}{2A} + \frac{x}{s}\right)J_c(x).$$

Here $H/2 = c/s = 1/(2A)$ and $2c/(a+b) = 1/s$. This proves the result, including the infinite tail beyond the polynomial's finite support. $\square$

Let $x_{a,b} = su_{a,b}$, and let x_B be the unique positive root of

$$G_\eta(x_B) = A, \quad G_\eta(x) = x\{I_1(2x) + \eta I_0(2x)\}. \quad (14)$$

The positive series makes G_η strictly increasing. Standard Bessel identities also give

$$G'_\eta(x) - 2G_\eta(x) = 2x(1-\eta)\{I_0(2x) - I_1(2x)\} + \eta I_0(2x) \geq 0. \quad (15)$$

The sign $I_0(2x) > I_1(2x) > 0$ follows from their integral representations. Hence $(\log G_\eta)' \geq 2$, which controls the inverse of G_η uniformly in η .

Corollary 6.3 (quantitative crossing bound). *We have $x_{a,b} \geq x_B$. If*

$$\delta_{a,b} = \frac{x_{a,b}^2}{2A} + \frac{x_{a,b}}{s} < 1,$$

then

$$0 \leq x_{a,b} - x_B \leq -\frac{1}{2}\log(1 - \delta_{a,b}). \quad (16)$$

Proof. At the exact crossing, Theorem 6.2 gives $A \leq G_\eta(x_{a,b}) \leq A/(1 - \delta_{a,b})$. Integrate $(\log G_\eta)' \geq 2$, which follows from (15). $\square$

6.2 Every growing distance from an edge

Theorem 6.4 (uniform crossing expansion). *Suppose $1 \leq a \leq b$ and $a \rightarrow \infty$, with no restriction on the relative growth of a and b . Put*

$$L = \log A, \quad C(\eta) = \frac{1}{2}\log(8\pi) - \log(1 + \eta), \quad B(\eta) = \frac{3 - \eta}{8(1 + \eta)}.$$

Uniformly over $b \geq a$,

$$\begin{aligned} 2\sqrt{ab}u_{a,b} &= L - \frac{1}{2}\log L + C(\eta) \\ &\quad + \frac{\frac{1}{4}\log L - \frac{1}{2}C(\eta) + B(\eta)}{L} + O\left(\frac{(\log L)^2}{L^2} + \frac{(\log a)^2}{a}\right). \end{aligned} \quad (17)$$

The same expansion and error order hold with $1 - p_{a,b}$ in place of $u_{a,b}$.

Proof. We have $a/2 \leq A \leq a$, $s \geq a$, and $0 < \eta \leq 1$. At $x = \log a$, the relative error in Theorem 6.2 tends to zero uniformly, whereas

$$\frac{G_\eta(x)}{A} \geq \frac{xI_1(2x)}{a} \rightarrow \infty.$$

The limit follows from the large-argument expansion of I_1 . Hence $x_{a,b} \leq \log a$ for all sufficiently large a , uniformly over b . Corollary 6.3 now implies

$$x_{a,b} - x_B = O((\log a)^2/a). \quad (18)$$

Write $z = 2x_B$. The usual large-argument expansions [17, Eq. 10.40.1], combined with $0 \leq \eta \leq 1$, give uniformly

$$I_1(z) + \eta I_0(z) = \frac{(1+\eta)e^z}{\sqrt{2\pi z}} \left\{ 1 + \frac{\eta-3}{8(1+\eta)z} + O(z^{-2}) \right\}.$$

Thus (14) becomes

$$z + \frac{1}{2} \log z = L + C(\eta) + \frac{B(\eta)}{z} + O(z^{-2}).$$

Initially this gives $z = L + O(\log L)$ uniformly. Substitution of $z = L + d + e/L$ then gives

$$d = -\frac{1}{2} \log L + C(\eta), \quad e = \frac{1}{4} \log L - \frac{1}{2} C(\eta) + B(\eta).$$

The residual is $O((\log L)^2/L^2)$, and the derivative of $z + \frac{1}{2} \log z$ is at least one. Combining this inversion with (18) proves (17). Finally,

$$2s \left(u - \frac{u}{1+u} \right) = O(su^2) = O((\log a)^2/s) = O((\log a)^2/a),$$

which proves the assertion about $1 - p_{a,b}$. □

In particular,

$$1 - p_{a,b} \sim \frac{\log a}{2\sqrt{ab}} \quad (a \rightarrow \infty, b \geq a), \quad (19)$$

uniformly in b . If $a = o(b)$, the more precise form is

$$2\sqrt{ab}(1 - p_{a,b}) = \log a - \frac{1}{2} \log \log a + \frac{1}{2} \log(8\pi) + o(1). \quad (20)$$

For a fixed bulk fraction $a/N \rightarrow \alpha \in (0, 1/2]$, the corresponding formula is

$$\begin{aligned} 2\sqrt{a(N-a)}(1 - p_{a,N-a}) &= \log N - \frac{1}{2} \log \log N + \log\{\alpha(1-\alpha)\} \\ &\quad + \frac{1}{2} \log(8\pi) - \log\{1 + 2\sqrt{\alpha(1-\alpha)}\} + o(1). \end{aligned}$$

We keep $2\sqrt{a(N-a)}$ on the left instead of $2N\sqrt{\alpha(1-\alpha)}$, since no rate of convergence of a/N is assumed. At the center this gives the constant $\frac{1}{2} \log(\pi/8)$ in Theorem 5.2. More precisely, setting $a = b = N/2$ gives $A = N/4$ and $\eta = 1$, and changing from $\log A$ to $\log N$ in Theorem 6.4 gives its inverse-logarithm term as well. No condition such as $a \gg \log^2 N$ is needed, since the Bessel argument at an edge is of order $\log a$ and the depletion error is $O(\log^2(a)/a)$ for every growing a .

6.3 Fixed bins and the Laguerre edge law

If a stays fixed while $b \rightarrow \infty$, the a letters R sit in a long string of L 's. A run of R 's away from the ends needs two switches, which gives a factor u^2 , and it can go in about b places. So what matters is $y = bu^2$, and the crossing is at $u \approx c_a/\sqrt{b}$.

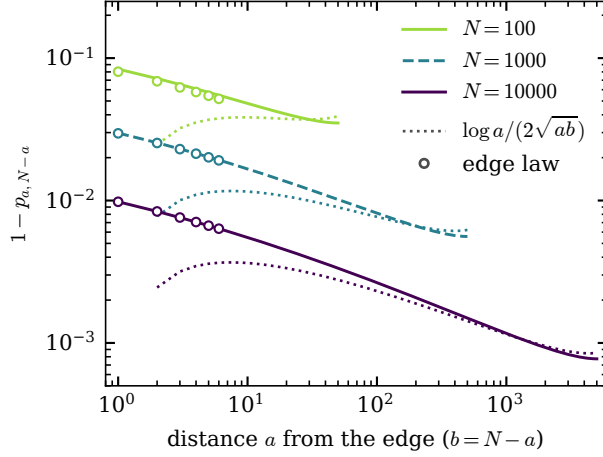


Figure 4: The threshold $1 - p_{a, N-a}$ of every bin against its distance a from the nearer edge. The dotted lines are the leading law (19), and the circles are the edge law (23) for $a \leq 6$. The right end of each line is the central crossing $1 - p_N$.

For a positive integer a , define

$$H_a(y) = \sum_{r=0}^{a-1} \binom{a-1}{r} \frac{y^{r+1}}{(r+1)!} = \frac{y}{a} L_{a-1}^{(1)}(-y), \quad (21)$$

where $L_{a-1}^{(1)}$ is the generalized Laguerre polynomial in its standard normalization. Let y_a be the unique positive root of $H_a(y_a) = 1$, and put $c_a = \sqrt{y_a}$. We have $y_1 = 1$, $y_2 = \sqrt{3} - 1$, and y_a decreases strictly with a , by coefficient comparison.

Theorem 6.5 (fixed-edge expansion). *For each fixed positive integer a , as $b \rightarrow \infty$,*

$$u_{a,b} = \frac{c_a}{\sqrt{b}} - \frac{1}{b} + \frac{\gamma_a}{b^{3/2}} + O_a(b^{-2}), \quad (22)$$

where

$$\gamma_a = \frac{1 + y_a + (y_a + y_a^2/2)H_a''(y_a)/H_a'(y_a)}{2c_a}.$$

Thus the second coefficient in the odds expansion is -1 for every fixed bin. For the switching probability,

$$1 - p_{a,b} = \frac{c_a}{\sqrt{b}} - \frac{1 + c_a^2}{b} + O_a(b^{-3/2}). \quad (23)$$

Proof. Put $\epsilon = b^{-1/2}$, $y = bu^2$, and $V_r = \prod_{j=1}^r (1 - j/b)$. With a fixed and $b \geq a$, the exact run polynomial can be written

$$\begin{aligned} S_{a,b}(u) &= 2\sqrt{y/b} \sum_r \binom{a-1}{r} \frac{y^r}{r!} V_r \\ &\quad + \frac{y}{b} \sum_r \binom{a-1}{r+1} \frac{y^r}{r!} V_r + y \sum_r \binom{a-1}{r} \frac{y^r}{(r+1)!} V_{r+1}. \end{aligned} \quad (24)$$

All these sums have bounded length. Expanding $V_r = 1 - r(r+1)/(2b) + O_a(b^{-2})$ gives, uniformly near $y = y_a$,

$$S_{a,b}(u) = H_a(y) + 2\sqrt{y} H'_a(y)\epsilon + \{(y - y^2/2)H''_a(y) - yH'_a(y)\}\epsilon^2 + O_a(\epsilon^3). \quad (25)$$

In particular, the second-order term uses the identity

$$\sum_r (r+1)(r+2) \binom{a-1}{r} \frac{y^{r+1}}{(r+1)!} = y^2 H''_a(y) + 2y H'_a(y).$$

The finite-product expression (24) extends analytically in (ϵ, y) near $(0, y_a)$, and $H'_a(y_a) > 0$. The implicit-function theorem therefore gives the unique crossing as

$$y = y_a + d_1\epsilon + d_2\epsilon^2 + O_a(\epsilon^3).$$

Matching coefficients in (25) gives

$$d_1 = -2c_a, \quad d_2 = 2 + y_a + (y_a + y_a^2/2) \frac{H''_a(y_a)}{H'_a(y_a)}.$$

Expanding $u = \epsilon\sqrt{y}$ proves (22). Expanding $u/(1+u)$ gives (23). $\square$

The next theorem shows that the edge law and the growing-bin law agree, without interchanging the two limits.

Theorem 6.6 (matching the edge and Bessel regimes). *For every positive integer a and every $x > 0$,*

$$0 \leq 1 - \frac{aH_a(x^2/a)}{xI_1(2x)} \leq \frac{x^2}{2a}. \quad (26)$$

Consequently, with $\ell = \log a$ and $C_0 = \frac{1}{2} \log(8\pi)$,

$$2\sqrt{ay_a} = \ell - \frac{1}{2} \log \ell + C_0 + \frac{\frac{1}{4} \log \ell - \frac{1}{2} C_0 + 3/8}{\ell} + O\left(\frac{(\log \ell)^2}{\ell^2} + \frac{(\log a)^2}{a}\right). \quad (27)$$

Proof. With $U_r = (r!/a^r) \binom{a-1}{r}$ as before,

$$H_a(x^2/a) = \frac{x}{a} \sum_{r \geq 0} \frac{x^{2r+1}}{r!(r+1)!} U_r.$$

Apply $0 \leq 1 - U_r \leq r(r+1)/(2a)$ and the identity

$$\sum_{r \geq 0} r(r+1) \frac{x^{2r+1}}{r!(r+1)!} = x^2 I_1(2x)$$

to obtain (26). The same root comparison as in Corollary 6.3 compares $\sqrt{ay_a}$ with the root of $xI_1(2x) = a$, with error $O(\log^2(a)/a)$. Invert that equation as in Theorem 6.4, now with $\eta = 0$ and $A = a$. Here $C(0) = C_0$ and $B(0) = 3/8$, which proves the expansion. $\square$

Figure 4 compares the exact thresholds with (19) and (23).

6.4 The moving boundary layer

The all-bin threshold theorem also describes the shape of a row at a prescribed switching probability. Let $v_N > 0$ be the chosen odds, so $p = 1/(1 + v_N)$, and define

$$\mathcal{L}_N = \#\{1 \leq k \leq \lfloor N/2 \rfloor : P_N(k) < P_N(0)\}.$$

By Proposition 6.1, these bins form an initial interval adjacent to the endpoint.

Corollary 6.7 (boundary-layer width). *Suppose*

$$T_N = \frac{1}{\sqrt{N} v_N} \rightarrow \infty, \quad \frac{\log T_N}{N v_N} \rightarrow 0.$$

Then

$$\mathcal{L}_N \sim T_N^2 \log^2 T_N = \frac{\log^2(1/(\sqrt{N} v_N))}{N v_N^2}. \quad (28)$$

In particular, if $v_N = N^{-\gamma}$ with $1/2 < \gamma < 1$, then

$$\mathcal{L}_N \sim (\gamma - \frac{1}{2})^2 N^{2\gamma-1} \log^2 N.$$

By symmetry, the total number of interior bins below the endpoint height is $2\mathcal{L}_N$ for all sufficiently large N .

Proof. Set $d_N = T_N^2 \log^2 T_N$. The assumptions give $d_N \rightarrow \infty$ and

$$\frac{d_N}{N} = \left(\frac{\log T_N}{N v_N} \right)^2 \rightarrow 0.$$

For each fixed $\varepsilon \in (0, 1)$, choose integers $a_{\pm} = (1 \pm \varepsilon)d_N + O(1)$. Theorem 6.4, or its leading form (19), gives

$$\frac{u_{a_{\pm}, N-a_{\pm}}}{v_N} \sim \frac{\log a_{\pm}}{2\sqrt{a_{\pm}N} v_N} \rightarrow \frac{1}{\sqrt{1 \pm \varepsilon}}.$$

Here $\log a_{\pm} \sim 2 \log T_N$ was used. Thus bin a_- lies below the endpoint height, while bin a_+ lies above it, for all sufficiently large N . Strict ordering then bounds $\mathcal{L}_N$ between these indices. Letting $\varepsilon \downarrow 0$ proves the result. Finally $\mathcal{L}_N = o(N)$, so the reflected intervals are disjoint and contain every bin below the endpoints. $\square$

7 Boundary conditions and a spatial crossing window

The binary model is also a zero-field Ising chain with free boundaries. For step signs $\varepsilon_i \in \{-1, 1\}$ its weight is proportional to $\exp(K \sum_{i=1}^{N-1} \varepsilon_i \varepsilon_{i+1})$, where $u = q/p = e^{-2K}$. The analogous periodic measure includes the last-to-first bond. Antal, Droz and Rácz [12] derive boundary-dependent limiting magnetization laws, and Dantchev and Rudnick [13] give exact free and periodic coefficients. Stepanyan et al. [14] study the periodic endpoint-adjacent and endpoint-center comparisons directly. So neither the periodic law nor the crossing question is new. Here we bound the discrete error uniformly and give a large- N expansion that refines their finite-size approximation.

Note that the periodic spin measure is not the same as wrapping the free Galton walk around a cylinder, which is the case in Remark 4.3. Write $R_{N,k}^F$ and $R_{N,k}^P$ for bin-to-endpoint ratios under free and periodic boundaries. Put $a = k$, $b = N - k$ and $s = \sqrt{ab}$. The classical cyclic count is

$$R_{N,k}^P(u) = \sum_{r=1}^{\min(a,b)} \frac{N}{r} \binom{a-1}{r-1} \binom{b-1}{r-1} u^{2r}. \quad (29)$$

Indeed, choose compositions of each letter count into r positive runs, choose a labeled position for the start of a distinguished run of one letter, and divide by its r possible choices. The normalization of the periodic measure cancels in the ratio. In particular $R_{N,1}^P = Nu^2$, giving the known adjacent threshold $K = (\log N)/4$. Every interior ratio is strictly increasing from zero to infinity in $u > 0$, and the ratios increase strictly toward the middle, except for the symmetric middle-bin tie. The coefficient comparison uses the same products $(a-j)(b-j)$ as the free case.

Theorem 7.1 (periodic relative error). *For $N \geq 2$, $1 \leq k \leq N - 1$, and $u > 0$, define*

$$T_{N,k}^P(u) = \frac{Nu}{\sqrt{k(N-k)}} I_1\left(2u\sqrt{k(N-k)}\right).$$

Then

$$0 \leq 1 - \frac{R_{N,k}^P(u)}{T_{N,k}^P(u)} \leq \frac{Nu^2}{2}. \quad (30)$$

Proof. Set $x = su$, $A = ab/N$, and $U_r = r! \binom{a-1}{r} / a^r$, $V_r = r! \binom{b-1}{r} / b^r$. The clipped-product inequality gives $0 \leq 1 - U_r V_r \leq r(r+1)/(2A)$, including the zero factors beyond support. After shifting the index in (29), the ratio to T^P is the weighted average of $U_r V_r$ with weights $x^{2r+1}/(r!(r+1)!)$. Their sum is $I_1(2x)$, while their $r(r+1)$ -weighted sum is $x^2 I_1(2x)$. Thus the relative deficit is at most $x^2/(2A) = Nu^2/2$. $\square$

We can guess the $\log 2$ in the next theorem. A cyclic word always switches an even number of times, but a free word can also start and end on different letters, and those words give the I_0 term in the free approximation. For large arguments $I_0 \approx I_1$, so near the crossing the free ratio is about twice the periodic one. Both grow like e^{Nu} , so the periodic crossing should need Nu to be about $\log 2$ larger.

Theorem 7.2 (periodic crossing and boundary shift). *Let u_N^P be the positive central root, and let $u_N^F = u_N$ be the free central root. Put $L = \log N$ and $c_P = \frac{1}{2} \log(\pi/2)$. Then*

$$Nu_N^P = L - \frac{1}{2} \log L + c_P + \frac{\frac{1}{4} \log L - \frac{1}{2} c_P + \frac{3}{8}}{L} + O\left(\frac{(\log L)^2}{L^2}\right), \quad (31)$$

$$N(u_N^P - u_N^F) = \log 2 + \frac{\frac{1}{4} - \frac{1}{2} \log 2}{\log N} + o(1/\log N). \quad (32)$$

Replacing Nu_N^P by $N(1 - p_N^P)$ preserves (31). In the Ising convention $K = \beta J$,

$$\beta_N^P J = \frac{1}{2}(\log N - \log \log N) + \frac{\frac{1}{4} \log \log N - \frac{1}{2} c_P}{\log N} + O\left(\frac{(\log \log N)^2}{(\log N)^2}\right). \quad (33)$$

Proof. At the middle, $2s = N + O(1/N)$. Evaluate Theorem 7.1 at $u = c \log N/N$ for fixed $c > 0$. Its relative error tends to zero, and the Bessel expansion makes the ratio tend to zero when $c < 1$ and to infinity when $c > 1$. Monotonicity therefore gives $Nu_N^P \sim \log N$. Writing $z = Nu_N^P$ and using $I_1(z) = e^z(2\pi z)^{-1/2}(1 - 3/(8z) + O(z^{-2}))$ in the estimate at the root yields

$$z + \frac{1}{2} \log z = L + c_P + \frac{3}{8z} + O(z^{-2} + L^2/N).$$

The odd- N argument change contributes only $O(L/N^2)$ to this logarithmic equation. Substitute $z = L + d + e/L$. The constant terms give $d = -\frac{1}{2} \log L + c_P$, and the $1/L$ terms give $e = 3/8 - d/2$. The derivative of the left side is at least one, so the residual yields the stated error. Since $N(u - u/(1+u)) = O(L^2/N)$, the switching-probability version follows. Subtract the free expansion, where $c_F = \frac{1}{2} \log(\pi/8)$ and the correction is $1/8$, to obtain (32). Finally $\beta J = -\frac{1}{2} \log u = \frac{1}{2}(L - \log z)$ gives (33). $\square$

Stepanyan et al. describe their formula [14, Eq. (16)], which is linear in $\log N$, as a finite-size approximation. Equation (33) gives the large- N behavior with a proved error term.

The factor e^{-2y^2} is not hard to see. If we move $yN/\sqrt{\log N}$ away from the middle, the Bessel argument is multiplied by $2\sqrt{k(N-k)}/N \approx 1 - 2y^2/\log N$. The argument is about $\log N$, so it goes down by about $2y^2$, and since the ratio grows roughly like e to the argument, it goes down by a factor of about e^{-2y^2} .

Theorem 7.3 (spatial crossing window). *Let $X \in \{F, P\}$ and let u_N^X be its exact central crossing. For fixed real t, y set*

$$u = u_N^X + \frac{t}{N}, \quad k_N(y) = \text{nint}\left(\frac{N}{2} + \frac{yN}{\sqrt{\log N}}\right),$$

where either nearest-integer convention may be used. Then, uniformly on compact sets of (t, y) ,

$$R_{N, k_N(y)}^X(u_N^X + t/N) \longrightarrow e^{t-2y^2}. \quad (34)$$

If $M_N^X(t)$ is the number of interior bins with probability strictly above the endpoint probability at that parameter, then

$$\frac{\sqrt{\log N}}{N} M_N^X(t) \longrightarrow \sqrt{2 \max(t, 0)}. \quad (35)$$

Proof. Put $L = \log N$, $\delta = k/N - 1/2$, and $\eta = 2\sqrt{k(N-k)}/N = \sqrt{1 - 4\delta^2}$. For the stated rounding and bounded y ,

$$\eta = 1 - \frac{2y^2}{L} + O\left(L^{-2} + \frac{1}{N\sqrt{L}}\right), \quad Nu_N^X = L + O(\log L).$$

Thus the Bessel argument at (u, k) equals $(Nu_N^X + t)\eta = Nu_N^X + t - 2y^2 + o(1)$, uniformly on the compact set. Use the free all-bin estimate and Theorem 7.1. Both have relative discrete error $O(L^2/N)$ here. The Bessel expansion $I_\nu(z) = e^z/\sqrt{2\pi z}(1 + O(1/z))$, for $\nu = 0, 1$, shows that the prefactors relative to their central values tend uniformly to one. The argument change contributes e^{t-2y^2} . Since the central exact ratio is one, this proves (34).

The interior ratios increase toward the middle. If $t \leq 0$, no interior bin is strictly above the endpoint at or below the central crossing. If $t > 0$, for any $0 < \epsilon < \sqrt{t/2}$ the bins at scaled distance less than $\sqrt{t/2} - \epsilon$ are above the endpoint, whereas those at distance $\sqrt{t/2} + \epsilon$ are below.

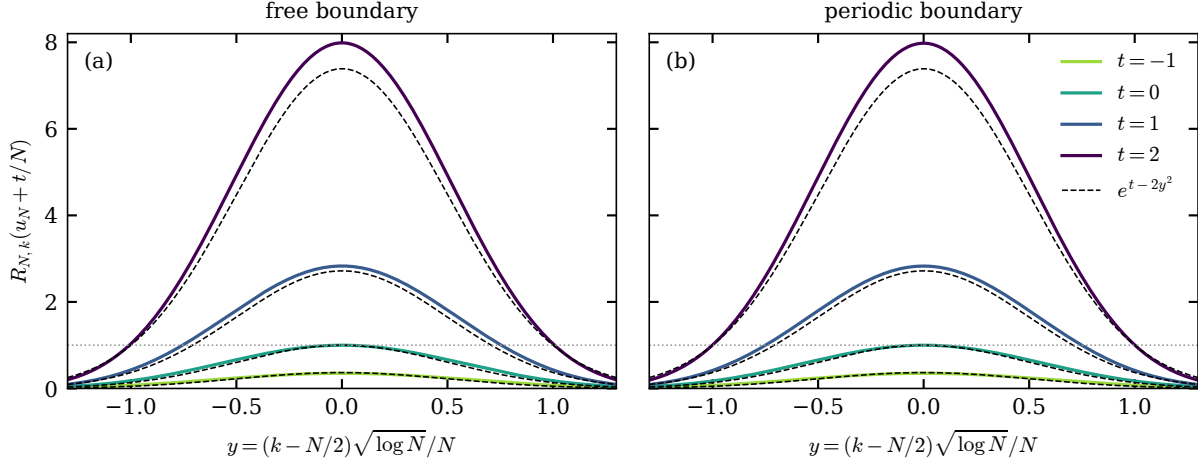


Figure 5: The exact ratios $R_{N,k}^X(u_N^X + t/N)$ at $N = 10^6$ for free (a) and periodic (b) boundaries, with the limit e^{t-2y^2} of Theorem 7.3 (dashed). Here $y = (k - N/2)\sqrt{\log N}/N$, and the bins above the dotted line are above the endpoint height. The convergence is slow, and at $t = 2$ and $y = 0$ the relative error is still 0.08.

Monotonicity excludes all more distant bins, even outside the compact interval on which the local limit was applied. Count the resulting interval, divide by $N/\sqrt{L}$, and let $\epsilon \downarrow 0$. Rounding changes the count by $O(1)$. $\square$

The window has width about $1/N$ in p and about $N/\sqrt{\log N}$ in space. A weak limit theorem would not give this, since it says nothing about a single bin, and in particular nothing about the endpoint. Figure 5 shows the exact ratios at $N = 10^6$.

8 Verification and scope

We also checked the results by computer, using Python with only the standard library. We compared the exact law with a dynamic program and with a list of every word, and the periodic run polynomial with a separate count of cyclic words. We also checked the coefficient identities and the order of the crossings, and evaluated the Bessel bounds and expansions to high precision. None of the proofs use these checks. **SUPPLEMENT.md** lists the scripts and the parts that are proved in Lean, and the output is in **data/**. The case of more than two directions is in a separate paper.

Data and code availability. The manuscript source, verification scripts, and recorded results are in the Problem 131 folder of <https://github.com/apemm/Kagey-Problems>.

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